ROOT CAUSE UNLOCKED · RECOVERY GUIDE

The Low Back and Hip *Recovery* Guide

The in-clinic tissue prep protocols, the movement evidence, and the honest supplement picture for low back pain, sciatica, and hip pain

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The Low Back and Hip Recovery Guide

BEFORE YOU START

Contents

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Some back pain is an emergency. Know these before anything else in this guide:

- New numbness in the saddle area (groin, inner thighs, the parts that would touch a bicycle seat), new trouble controlling bladder or bowels, or progressive weakness in both legs. This pattern can mean cauda equina compression. **Emergency room, now, not tomorrow.**
- Back pain with fever, back pain after significant trauma, or new back pain if you have a history of cancer, IV drug use, or a suppressed immune system. Same-day medical evaluation.
- A fall followed by groin or hip pain and inability to bear weight. That is fracture-until-proven-otherwise territory.
- Night pain that is unrelenting regardless of position, week after week, deserves a physician visit rather than a foam roller.

The tool safety rules, from the same sheets we use in clinic: do not work over an acute injury in its first 48 to 72 hours; reduce pressure over inflamed areas; skip percussion devices entirely if you have a pacemaker or other implanted electronic device; avoid aggressive pressure over varicose veins or compromised skin; be cautious if you take blood thinners. Start light. On the manual tools, the foam roller, roller stick, ball and trigger point tool, where the pressure is your own body weight, pain must not exceed 6 out of 10: uncomfortable but tolerable. **On a powered percussion device the rule is different and stricter.** Follow the manufacturer's instructions, which say to use it without producing pain or discomfort and to stop immediately if you feel any, apart from light muscle soreness. I am not asking you to push a powered tool to a number. Sharp or shooting pain means back off, whatever the tool.

Nothing here is a reason to stop a prescription or skip a recommended evaluation.

Why one guide covers the low back and the hip

The tools sort by region, and so does the biology. The low back and hip share their major movers (glutes, hip flexors, hamstrings, adductors), share load between them with every step, and irritate each other when either stops doing its share. Half of what presents as “low back pain” carries a hip contribution, and lateral hip pain is routinely mislabeled, including by the people who have it. This guide covers the region: low back pain, sciatica, and hip pain, including the gluteal tendinopathy that most “hip bursitis” actually is.

If you were told you have... (your diagnosis, translated)

Diagnostic labels for this region have changed, and most readers arrive carrying an older one. Here is where yours lives in this guide:

- **“Trochanteric bursitis,” “hip bursitis,” or “greater trochanteric pain syndrome”:** the lateral hip pain territory. The research now treats most of it as gluteal tendinopathy rather than an inflamed bursa, and the LEAP trial in part two is your evidence.
- **“Sciatica,” a “pinched nerve,” or a “bulging disc”:** the sciatica material in part two, plus the companion Sciatica ebook. A disc bulge on imaging is common in pain-free backs; the finding and the pain are not the same thing.
- **“Degenerative disc disease” or “arthritis of the spine”:** labels for common age-related change that sound far scarier than their prognosis. The low back program applies.
- **“Piriformis syndrome”:** a community label for deep buttock pain; the glute and hip work in this guide covers that territory.
- **“Spinal stenosis”** (leg symptoms when walking that ease when you sit or lean forward): this deserves its own clinical evaluation before assuming this guide’s program fits, especially past sixty.

The one idea that makes this kit honest

Here is the sentence that separates this guide from most of what sells in this category: **the tools buy you a window, and the movement spends it.**

The evidence for massage tools is real but modest, and it is short-term: foam rolling used as a warm-up produces small improvements in flexibility and sprint performance (PMID 31024339), and massage guns improve short-term range of motion, flexibility, and recovery measures, while doing nothing for strength or performance, and are actually not recommended immediately before demanding strength or balance work (PMID 37754971). That last point changes the order of your session rather than the contents of it, so let me be specific instead of leaving you to reconcile it: the rolling, the stretches, the heat and the ball work can all run straight into your movement work, and the percussion massager should not. Use it after a session, on a rest day, or leave roughly half an hour before you load. Heat has the same honest profile, and one finding worth reading twice: moderate evidence shows heat wrap therapy gives a small short-term reduction in low back pain and disability, **and adding exercise to the heat further reduces pain and improves function** (PMID 16437495). The tools prepare tissue and calm symptoms so that moving feels possible. The movement is the treatment. Everything in this guide is sequenced around that fact, which is exactly how these same protocols are used before treatment in our clinic.

Part one: the tissue prep protocols, exactly as used in clinic

These are the same protocols patients receive at Root Health, tools, targets, and doses unchanged. Do them before your movement work, not instead of it. Respect the pressure ceilings above, manual versus powered, throughout.

One timing rule carries over from the section above, and it is the only place the order matters: the percussion massager is the one tool not to use immediately before strength work. Everything else here can run straight into your session.

The low back sequence

Heat and vibration back wrap, set to vibrate and heat, worn while performing the following:

- **Cat-camel stretch:** on hands and knees, arch the back up, then let it sag. 1 set of 10 repetitions.
- **Hamstring stretch with stretch strap:** on your back, strap around the foot, leg straight up with a soft knee. 1 set of 10 reps each side.
- **Ball bounce:** seated on an exercise ball, bounce up and down, then side to side, palms up throughout. 1 set of 10 each.

Massage ball or foam roller: lie on your back, place it on the glute muscles, and move the leg up and down. Work each side.

Massage gun, round or flat head: run the device over the TFL and hip flexors for 1 to 2 minutes each side.

Massage ball: place over the abdomen (psoas territory) and rock back and forth, then up and down. 20 to 30 seconds each side. This one is unusual and it is deliberate: the psoas connects the lumbar spine to the leg, and it rarely gets touched by anything.

Lumbar support device or rolled towel: lie on your back with the device under the lower back, knees bent to start (straighten if comfortable), and rest 30 to 60 seconds.

Massage ball or foam roller on the calf: find a sore spot, then move the foot up, down, left, right, and in circles. 2 to 3 repetitions per area. The calves are part of the chain that decides how your gait loads your spine.

The hip sequence

Massage gun, round or flat head:

- TFL and hip flexors, 1 to 2 minutes.
- All three glute muscles (maximus, medius, minimus), 1 to 2 minutes per side.
- Adductors (inner thigh), gently, 1 minute per side.

Massage ball or foam roller:

- Glutes: lie on your back with the sphere on the glutes and move the leg up and down, or sit on the roller and search for tender spots. 2 to 3 minutes per side.
- Hip flexor (psoas, iliacus): roll over the front of the hip, 1 to 2 minutes per side.
- IT band: side-lying, roll the outer thigh from below the hip to above the knee, 1 to 2 minutes per side.

Stretches: cat-camel, 1 set of 10. Hamstring stretch with the strap, 1 set of 10 each side. Modified pigeon pose for the deep hip rotators, 20 to 30 seconds per side, only if comfortable.

One caution specific to lateral hip pain: if your pain lives on the outside of the hip and hurts to lie on, go easy on direct sustained pressure there (including sitting on the sphere and aggressive IT band rolling). Compressive load over an irritated gluteal tendon can flare it. Treat above and around it, and let the loading program in part two do the heavy lifting.

Part two: the movement evidence, which is the actual treatment

This is the least glamorous and best-supported section of the guide. Nobody profits from telling you to walk, which, given how this guide reads funding everywhere else, is worth something. The 2026 hip review below lists its funding as none at all.

For chronic low back pain, exercise works, and resting does not. The Cochrane review of exercise therapy for chronic low back pain found moderate-certainty evidence that exercise is effective for pain compared with no treatment, usual care, or placebo, and that exercise probably beats advice or education alone (PMID 34580864). Which style of exercise matters far less than doing one: the review found no clear winner between approaches, so the best program is the one you will actually repeat. For a new episode of acute back pain, the advice with moderate-quality evidence behind it is to stay active rather than rest in bed (PMID 20556780). Bed rest is not treatment. It is deconditioning with a mattress.

For sciatica, time is on your side more than most people are told. In the randomized trial comparing early surgery against prolonged conservative care for severe sciatica, early surgery relieved leg pain faster, but disability outcomes were no different across the first year, and only 39 percent of the conservative group ended up needing surgery at all (PMID 17538084). At five years, outcomes between the groups were still equivalent (PMID 23793663). Severe or progressive weakness, or any of the red flags on page one, changes that calculus immediately. Short of those, a patient and progressive rehab approach is not the timid option; it is the evidence-based one, with surgery remaining available if genuinely needed.

For hip osteoarthritis, the honest answer got harder in 2026, and you should have it anyway. The Cochrane review on this question was rebuilt in July 2026, and it is the most sobering piece of evidence in this guide. Across 18 studies and 1,368 participants, exercise compared against a placebo or attention control may have little to no effect on pain, and may improve physical function only slightly, both on low-certainty evidence. Compared against no treatment or usual care, exercise probably does reduce pain and improve function, but the reviewers say plainly that the size of those improvements is unlikely to be clinically meaningful. And where exercise was added on top of another treatment, it probably made little to no difference to pain or function at all (PMID 42486497).

That is a smaller and less certain benefit than the previous edition of this review reported. The hip is not the knee, the evidence base here is much thinner, and I would rather hand you the current number than a more flattering one that has been superseded.

So why is exercise still the first thing I do for an arthritic hip, and the first thing every guideline recommends? Three reasons, and none of them require overselling. The alternatives for hip osteoarthritis are worse: injections, medications with real organ costs, and eventually a replacement. Exercise in that same review probably reduced the risk of adverse events slightly rather than raising it, which almost nothing else on the list can claim. And the trial averages you are reading are built from people who did the program and people who did not, which is precisely why the rest of this part is written to get you to actually do it. A small average benefit with no downside, in a condition where the other options carry real ones, is still the right first move. It is just not the miracle the internet sells, and you deserve the real number.

For lateral hip pain (gluteal tendinopathy), the trial worth knowing beat the injection. In a three-arm randomized trial published in the BMJ, fourteen sessions of load-management education plus exercise produced higher rates of global improvement and lower pain at eight weeks than either a corticosteroid injection or a wait-and-see approach, and at 52 weeks the education-plus-exercise group still reported better global improvement than the injection group (PMID 29720374). If you have been told your outer-hip pain is “bursitis” and offered a needle first, this trial is the conversation-changer: the boring program of progressive loading and not compressing the tendon (less leg-crossing, less hanging on one hip, side-sleeping with a pillow between the knees) outperformed it.

The program

Start with the finding that takes the pressure off. The Cochrane review of exercise for chronic low back pain looked for a winning style and did not find one (PMID 34580864). No approach beat the others. Read that as permission rather than as a disappointment: you are not one correct exercise away from your result, and a boring program you repeat for three months outperforms a perfect one you abandon in week two. If you can only do part of what follows, do that part.

The core, for everybody in this guide

1. **Walk most days.** Start at a distance that is comfortable today, and add roughly ten percent a week. Hold at the same distance for a week whenever a week feels hard. This is the single item with the broadest support in the whole guide and it costs nothing.
2. **Strength, two or three times a week, on non-consecutive days.** Two to three sets of 8 to 12 repetitions, slow enough that you could stop at any point in the range.
3. **Hinge well.** Learn to bend from the hips with a stiff-but-not-rigid spine, and use it for everything you pick up. This is a skill, not an exercise, and it pays out every day.
4. **Do not rest it.** For a new episode, staying active beats bed rest on moderate-quality evidence (PMID 20556780). Move at whatever size you can.

Start here: pick the one that sounds like you

If your picture is chronic nonspecific low back pain:

- Glute work is your priority: bridges, then single-leg bridges, plus standing or side-lying hip abduction.
- Add a hinge pattern you can load, and the cat-camel from part one as daily movement rather than as treatment.
- Add walking, which for this picture is not the warm-up. It is the treatment.
- Expect a sawtooth, not a straight line. Good weeks and bad weeks inside an overall trend upward is what recovery actually looks like here.

If your picture is sciatica (leg pain, sometimes with numbness or tingling, often worse than the back pain itself):

- Time and load are on your side. The trials say most people get there without surgery, and that waiting does not cost you the option (PMID 17538084, PMID 23793663).
- Find the directions that centralise your symptoms, meaning the ones that pull the pain out of the leg and back toward the spine, and favour those. Moving pain toward the midline is progress even when the intensity has not changed yet.

- Walk in short, frequent bouts rather than one long one.
- **Progressive weakness, a foot that slaps or catches, or any saddle-area or bladder change overrides all of this and needs assessment now, not at six weeks.**

If your picture is an arthritic hip:

- Sit-to-stand from a chair, chair height as your difficulty dial, plus hip abduction and bridges.
- Stationary cycling or pool work if weight-bearing is the limit. The 2026 review found no evidence that one exercise type beats another, so pick what you will keep doing.
- Read the honest section above before you set expectations. The average benefit here is small. Consistency and stacking it with weight management and walking is how you get above that average.

If your picture is lateral hip pain (outside of the hip, hurts to lie on that side, often called bursitis):

- **This is the pathway where the evidence is strongest and most specific in this entire guide.** Education plus exercise beat a corticosteroid injection and beat waiting, at eight weeks and still at 52 weeks (PMID 29720374). Fourteen supervised sessions was the trial's dose, which tells you this is a months-long project rather than a fortnight's.
- The load-management half matters as much as the exercise half, and it is mostly about not compressing the tendon: less leg-crossing, less standing hanging on one hip, and a pillow between the knees when you sleep on your side.
- Progressive hip abduction loading is the exercise half. Start isometric, holding rather than moving, if the tendon is angry.
- Go easy with the tools directly over the painful spot, per the caution in part one.

How to make it harder

Progress one thing at a time, roughly every one to two weeks, and only when the current version is comfortable: more repetitions first, up to about 12; then a harder version of the movement; then added load; then more sets. If you add difficulty and the next two sessions feel worse, go back one step for a week. That is the method working, not a setback.

How to know it is going right, and when to back off

The soreness rule. Discomfort during and after this work is expected and is not damage. The test is the next morning. Soreness that has settled by then is a normal training response and you carry on. Soreness that is still there, or worse, means the last session was too much, so drop back to the version you handled comfortably.

Reduce the load if a flare lasts more than a couple of days, or if pain starts waking you at night.

Stop and get examined for anything in the warning list at the top of this guide, for leg symptoms that are spreading or intensifying, for new weakness, or for pain trending worse across weeks rather than better.

Give it twelve weeks before you judge it, and check in at six. Six weeks of honest effort with nothing to show is a reason to be examined rather than a reason to push harder. Track it, and let the trend rather than any single day judge the program.

When to get evaluated instead of self-treating

Beyond the emergencies on page one: pain that is not trending better by six weeks of honest effort, leg symptoms that are spreading or intensifying, new weakness (a foot that slaps or catches, a knee that buckles), or a flare pattern you cannot connect to load. Chiropractic, physical therapy, and medical evaluation are complements here, not rivals, and a clinician who examines you beats any guide that cannot.

Part three: the supplement half, honestly graded

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

So when this guide says "no evidence," it will tell you which of three things it means: *tested and failed* (a real trial ran and the answer was no), *never properly tested* (nobody could afford the trial, so absence of evidence is not evidence of absence), or *tested small and won* (real but modest trials, usually measuring markers or short-term pain, because that is what a small budget can afford). The asymmetry earns nutrients humility, not automatic credit: small nutrient trials carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. One rule, both directions, and I apply it to the supplement industry below exactly as hard as I apply it to drug companies.

Sort the tissue before you buy anything

This is the step that decides whether a supplement has any chance of helping you, and almost nobody takes it. “Low back and hip pain” is not one problem. It is at least three different tissues failing in different ways, and they do not want the same things. The aisle sells one joint capsule to all three of you.

The joint is the arthritic hip: a worn, irritated bearing surface in a joint that has turned inflammatory. That is the tissue nearly all joint-supplement research was actually done in, so for that group the evidence is borrowed only slightly.

The tendon is lateral hip pain, the ache on the outside of the hip that hurts to lie on, which you were probably told was bursitis. That is gluteal tendinopathy, and a tendon is a rope of collagen, not a bearing surface. It follows completely different rules, and it is the group most easily failed, because it gets lumped in with nonspecific back pain and told that nothing works.

The spine and its muscles is nonspecific low back pain, where the honest answer stays thin and I am not going to dress it up.

One rule holds across all three, and it is why this section sits behind the movement section rather than in front of it. Every trial worth citing below layered its supplement on top of a loading program. Nutrition supplies raw material. Load is what tells the body where to put it. A supplement taken instead of the program is money spent on a signal your body has no reason to act on.

The spine: tested and failed, said plainly

Glucosamine for low back pain. A double-blind randomized trial of 250 patients with chronic low back pain and degenerative lumbar changes found six months of glucosamine produced no reduction in pain-related disability versus placebo, at six months or at one year (PMID 20606148). If you take glucosamine for your back, that is the trial to know.

Vitamin D for low back pain. The systematic review and meta-analysis conclusion is direct: not more effective than placebo or other interventions, and “the prescription of vitamin D for LBP cannot be recommended” on current evidence (PMID 29565945). Correcting a genuinely deficient vitamin D level still matters for bone and general health, and if you have a measured deficiency you should fix it on those terms. Expecting it to treat back pain is the part the evidence does not support. Test, then treat what the test shows, rather than treating a symptom with a nutrient that has been tried for exactly that and did not deliver.

For nonspecific low back pain, there is still no supplement with trial support worth your money, and I will not invent one. What I will do, instead of ending the paragraph there, is point you at the two things in this section that do apply to you regardless of which tissue you have: the protein floor and the load. Both are further down, both are free, and both outperform anything you would have bought here.

The arthritic hip: borrowed evidence, labeled as borrowed

The supplements with real positive signals in this territory were tested in **osteoarthritis**, overwhelmingly knee osteoarthritis. That transfers reasonably to an arthritic hip, which is the same disease in a different joint, and poorly to nonspecific low back pain, and this guide refuses to blur that line.

The systematic review and meta-analysis of dietary supplements for osteoarthritis, covering 20 supplements across 69 trials, found Boswellia serrata extract and curcumin among seven showing large and clinically important short-term pain reductions, while rating the overall quality of that evidence very low. At medium term only two supplements held a clinically important effect on pain, green-lipped mussel extract and undenatured type II collagen, and at long term none did (PMID 29018060). The Boswellia-specific meta-analysis of seven trials and 545 patients found meaningful short-term improvement in pain, stiffness and function, and concluded a fair trial runs at least four weeks (PMID 32680575). The same review found glucosamine and chondroitin either ineffective or showing small and arguably clinically unimportant effects even for osteoarthritis, which pairs with the back pain null above.

Curcumin has a delivery problem the label will not tell you about, and it decides whether you wasted your money. Plain curcumin is absorbed poorly. The osteoarthritis trial that showed a real effect used a specifically engineered high-absorption preparation delivering 180 mg of curcumin daily for eight weeks, measured at roughly 27 times the absorption of ordinary curcumin powder, and the patients on it had less pain and needed significantly less anti-inflammatory medication than placebo (PMID 25308211). The number that matters is not the milligrams of turmeric on the front of the bottle, it is whether the product uses a documented absorption-enhanced form. A large dose of a poorly absorbed powder is not a bigger dose, it is a more expensive one.

Oral hyaluronic acid is genuinely promising rather than proven. In a twelve-month double-blind placebo-controlled trial, 200 mg daily alongside daily quadriceps strengthening in both groups improved osteoarthritis symptom scores more than placebo, but reached significance only in subjects aged 70 or younger and only at the second and fourth months (PMID 23226979). That is sixty people, age-conditional, and industry-funded. If you are under 70 with an arthritic hip and already doing the strengthening, 200 mg daily for three months is a defensible experiment, and it is not a reason to skip the strengthening that both groups in that trial were doing.

Lateral hip pain: you have a tendon, and that changes everything

If your pain lives on the outside of the hip and hurts to lie on, stop reading the arthritis section. You have a collagen problem, and collagen has its own biology, its own timing, and its own evidence.

Start with what a tendon actually is. Roughly three quarters of the dry weight of tendon is type I collagen, a rope of protein wound from repeating glycine, proline and lysine. To build it, your body chemically modifies proline and lysine after the chain is assembled, and the enzymes that do that, prolyl hydroxylase and lysyl hydroxylase, cannot function without vitamin C and iron. The enzyme that then cross-links those ropes into something that carries load, lysyl oxidase, requires copper. Manganese and zinc are required elsewhere on the same assembly line. This is textbook connective tissue biochemistry rather than a supplement claim, and it is why a collagen program that supplies protein without supplying cofactors is doing half the job.

Now the timing, which is the part almost nobody gets right. Collagen synthesis in tendon is driven by mechanical load and rises in the hours after you load the tissue, so the useful question is not "should I take collagen" but "is the raw material in my bloodstream when my tendon is being told to rebuild." In the study that established this, eight healthy men took vitamin C-enriched gelatin an hour before six minutes of rope skipping, and the 15 gram dose doubled a blood marker of new collagen formation (PMID 27852613). A later dose-response trial found 30 grams of hydrolyzed collagen with 50 mg of vitamin C an hour before heavy resistance exercise produced more collagen synthesis than 15 grams, while **15 grams was not statistically different from nothing at all** (PMID 38007183). Most products on the shelf, and most people taking them, are dosed at or below the amount that failed to beat placebo. Run it at a dose the research supports, about an hour before the session, not at bedtime.

Here is the ceiling on that claim, and I would rather you hear it from me. A blood marker is not a healed tendon. When an independent group tested whether collagen peptides change tendon structure, giving 15 grams daily through fifteen weeks of resistance training with MRI endpoints, they found no advantage over placebo on tendon size, stiffness or mechanical properties. Both groups improved, and the training did the work (PMID 37436929). The most recent independent systematic review, nineteen studies and 768 participants, found small statistically significant effects and graded the tendon findings specifically as very low certainty (PMID 39060741). A positive trial does exist in a population close to many of you: 139 athletic adults with activity-related knee pain took 5 grams of bioactive collagen peptides daily for twelve weeks and reported significantly less activity-related pain than placebo (PMID 28177710). That trial carries a correction notice and one of its authors works for a collagen research institute, which is exactly the sponsorship pattern the Cochrane review at the top of this section warns about. I cite it anyway, labeled, because pretending it does not exist would be its own dishonesty.

So the fair summary is this. The mechanism is real and well described. The acute response is real and dose-dependent. Symptom trials are positive but small and mostly industry-adjacent. The one hard structural endpoint tested independently came back null. That is a cheap, low-risk experiment with honest odds, not a treatment with proven structural benefit, and anyone selling it as the latter is ahead of the evidence.

The cofactor half is the smarter half. If substrate is only useful when the assembly line is staffed, the cofactors matter as much as the grams, and they are what consumer products ignore. A practitioner formula supplying vitamin C, copper, manganese, zinc and the amino acids proline and lysine together is built on real biochemistry rather than a marketing story. Two honest notes. Its amino acid content is modest, typically around 150 mg each of proline and lysine, a rounding error next to the 15 to 30 grams used in the collagen trials, so it complements a collagen powder rather than replacing it. And these formulas often contain iron, which is a genuine cofactor here, but most men and most postmenopausal women should not take supplemental iron without a documented reason. Iron is one of the few nutrients where routinely taking more than you need is actively harmful.

Stage matters more than product, and this is the most useful paragraph in the section. In a three-arm randomized trial of 59 patients with Achilles tendinopathy, a supplement of mucopolysaccharides with type I collagen and vitamin C added benefit specifically in patients with early, reactive tendinopathy, and added nothing once the tendon had reached the degenerative stage with established matrix and vascular change (PMID 28053674). That trial was run in an Achilles, not a gluteal tendon, so treat the specific result as borrowed. What transfers is the principle: an early, angry, reactive tendon is still responsive tissue, and that is the window where nutritional support appears to add something, while a long-standing degenerative tendon has structural change that a capsule did not reverse. This is why "how long has this been going on" is one of the first questions I ask, and it should change what you spend money on. Weeks old, the supplement side is worth running. Two years old, put the money and the attention into the progressive loading program in part two, because that is what was actually shown to work at that stage, and in lateral hip pain specifically it outperformed a corticosteroid injection at 52 weeks (PMID 29720374).

The floor under all three tissues

The interventions with the best evidence in this entire section are not sold in the supplement aisle.

Protein is the one number worth knowing. The meta-analysis and meta-regression across 49 studies and 1,863 participants found protein supplementation meaningfully improves gains from resistance training, with benefits plateauing around 1.6 grams per kilogram of body weight per day (PMID 28698222). For a 180 pound person that is roughly 130 grams daily, and most people rehabbing an injury eat well under it. Collagen is a poor quality protein in isolation, low in the essential amino acids muscle needs, so treat a collagen dose as a targeted addition to your protein intake and never as your protein intake.

One disclosure on that paper, because I said I would run this rule in both directions. It carries a later correction noting that one of its authors sat on a sports supplement manufacturer's advisory board while it was written. The correction changed none of the data and the 1.6 figure stands, but holding the supplement industry to the standard I hold drug companies to means saying so when the study behind a number I like has a commercial tie on it, and not only when the number is one I dislike.

Sleep is when repair actually happens, energy sufficiency keeps repair funded, and at a weight-bearing joint any sustainable progress on excess load is mechanically meaningful with every step. Those outrank everything in a capsule and they earn me nothing.

What I would actually run, and how to judge it

Pick the tissue, not the product. Arthritic hip: an absorption-enhanced curcumin preparation or a Boswellia extract for eight to twelve weeks, optionally with 200 mg of oral hyaluronic acid daily if you are under 70, on top of the strengthening program. Lateral hip tendon pain: 15 to 30 grams of gelatin or hydrolyzed collagen with vitamin C about an hour before your loading sessions, cofactors covered, for twelve weeks, and weight it toward the early-and-reactive case rather than the long-standing one. Nonspecific low back pain: skip the aisle, hit the protein floor, and spend the money on the loading program instead. Underneath all three, 1.6 grams of protein per kilogram, sleep, and enough food.

Then judge it. Set a start date and an end date before you buy anything, change one thing at a time, and write down what you are measuring. Everything here is a defined trial with a stop date, not a subscription. If twelve weeks of honest effort produced nothing, that is a real answer worth having, and it sends you back to the loading program and to an examination instead of to the next bottle.

Interactions, and the ones people miss

Curcumin and Boswellia can interact with blood thinners, and curcumin at supplement doses can bother reflux-prone stomachs. Clear either with your prescriber if you take anticoagulants. Do not add supplemental iron, including inside a combination formula, without a reason to. Proprietary blends hide the dose of each ingredient behind a single total, which makes it impossible to know whether you are getting a researched amount of anything, so prefer products that disclose amounts. And whichever source you use, look for independent third-party testing, because the manufacturing side of this industry is even less regulated than the marketing side.

Curcumin and your liver. The National Center for Complementary and Integrative Health warns that the high-absorption curcumin formulations, the ones engineered so you absorb more of it, may harm the liver, and that liver damage has been reported in some people who took them. That warning applies to the product I name below, because high absorption is the entire reason I name it. The problem appears to be uncommon, and I would still rather you heard it from me than not at all. Stop the curcumin and call me if you notice unusual fatigue, nausea, loss of appetite, dark urine, or any yellowing of the skin or eyes. Do not take turmeric or curcumin supplements if you are pregnant, which the same source considers possibly unsafe. And treat eight weeks as a checkpoint rather than an open-ended course: we look at whether it actually helped and decide together whether to keep going.

Nothing in this section substitutes for being examined, and none of it applies to pain that is getting worse, waking you at night, or accompanied by any of the warning signs at the top of this guide.

Track it, so the program can be judged by data

The free Root Cause Unlocked tracker has an MSK mode built for exactly this guide: daily function, range of motion, load tolerated, and flare days, alongside the basics. Eight weeks of that turns “I think it might be helping” into an answer, and it makes any clinician visit dramatically more useful.

> ***Free tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=msk-lbhip-recovery***

Dr. Seay recommends: the exact products

The protocols in this guide are written generically on purpose, because the technique matters and the brand mostly does not. A rolled towel works. This section is the other half of that promise: if you would rather not stand in an aisle guessing, here is exactly what I would hand you, why that specific item and not a cheaper-looking equivalent, and where to get it. Read the disclosure at the end before you buy anything, because I want you deciding with your eyes open.

On the supplement side, through my dispensary

Everything below is available through the Root Health Fullscript dispensary. Search the product name exactly as written.

> ***us.fullscript.com/welcome/drseay***

For the arthritic hip: Theracurmin HP, by Integrative Therapeutics. Two capsules daily for eight weeks. I name this one specifically because it is the actual preparation used in the osteoarthritis trial discussed above, at the actual dose that trial used: two capsules deliver 180 mg of curcumin in the water-dispersible form measured at roughly 27 times the absorption of standard curcumin extract (PMID 25308211). This is the clearest case in this guide where the brand genuinely matters, because ordinary curcumin powder at the same milligram count is not the same product and did not produce those results.

Before you start this one, read the liver and pregnancy caution in the interactions section above. It applies specifically to high-absorption preparations, which is what this is.

For collagen cofactors, if lateral hip tendon pain is your picture: Collagenics, by Metagenics. Three tablets daily. This is the assembly line rather than the raw material: vitamin C, copper, manganese, zinc and the amino acids proline and lysine in one formula, which is the cofactor set the biochemistry section above describes. Read that section's iron caution first. Collagenics contains 6 mg of iron, and if you are a man or a postmenopausal woman without documented low iron, ask me or your prescriber before starting it rather than after.

For an early, reactive tendon: SynovX Tendon and Ligament, by Xymogen. Two capsules daily. This supplies the mucopolysaccharide and type I collagen combination with vitamin C tested in the tendinopathy trial above, where it added benefit in early-stage reactive tendinopathy and not in long-standing degenerative tendons (PMID 28053674). That trial was run in an Achilles, so match it to your stage rather than assuming it was proven in a hip. If your lateral hip pain is years old, spend the money on the loading program instead, and I would rather tell you that than sell you a bottle.

For collagen substrate: a plain hydrolyzed collagen or gelatin powder, 15 to 30 grams, with vitamin C. Here the brand genuinely does not matter and I will not pretend otherwise. What matters is the dose and the timing: aim toward the upper end of that range, take it about an hour before your loading session, and make sure vitamin C is on board. An unflavored bulk powder is the honest buy. Do not pay a premium for a proprietary blend delivering two grams.

If nonspecific low back pain is your picture, buy none of the above. The evidence does not support it and I am not going to sell you something to feel thorough. Put the money toward protein and toward the loading program.

Slots I have deliberately left open rather than filled: a Boswellia extract, an oral hyaluronic acid at 200 mg, and vitamin D. The Boswellia trials pooled several different extracts, so there is no single trial-matched product to point at the way there is with Theracurmin, and vitamin D should follow a test result rather than a guide. Ask me at your visit and I will pick these to fit you rather than guessing in print.

On the tool side

The protocols name categories, not brands, and every one can be done with inexpensive equipment. Any dense foam roller does the rolling work, a vibrating roller just adds vibration. A massage ball or lacrosse ball substitutes for a vibrating sphere. A rolled towel approximates the lumbar support device closely enough to try before you buy one. The percussion massager and the heat-plus-vibration wraps are the two items without a good household substitute, and they are also the most expensive, so they are the ones worth thinking hardest about.

If you want the specific equipment I use in the clinic, these are the links:

> ***Percussion massager, vibrating roller, vibrating sphere and heat-plus-vibration wraps:*** *[[HYPERICE_AFFILIATE_LINK]]* ***Foam roller, massage ball, stretch strap, lumbar support and exercise ball:*** *[[GENERAL_TOOLS_AFFILIATE_LINK]]*

Disclosure: Root Health earns a portion of dispensary purchases, and the tool links above are affiliate links, which means we may earn a commission if you buy through them at no additional cost to you. Nothing in this guide was included, excluded, ranked, or dosed based on what it earns. The generic substitutions listed above are real substitutions and I would rather you use a rolled towel than skip the work, and the highest-value items in this guide, walking, glute strengthening, protein, and sleep, earn us nothing at all.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your history, your exam, and your actual loading tolerance.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=msk-lbhip-recovery***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=msk-lbhip-recovery***

The companion ebooks go deeper on each condition in this region: *The Complete Patient's Guide to Low Back Pain*, *The Complete Patient's Guide to Sciatica*, and *The Complete Patient's Guide to Hip Pain*.

References

References 1 to 14 were verified against live PubMed records on 2026-08-09. References 15 to 23, added with the rebuilt supplement section, were verified against live PubMed records on 2026-09-04. Both passes included a publication type and corrections check for retractions and errata. On 2026-09-05 every reference was additionally screened for superseded editions, which is how reference 4 was moved to the 2026 Cochrane update; no other reference in this guide is superseded. That same screen surfaced a previously undisclosed correction on reference 23, noted there. Verification detail is documented in `Recovery-Guide-Citation-Ledger.md`.

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Educational content only. Not medical advice. Not a diagnosis. Not a substitute for examination. The emergency signs at the top of this guide outrank everything else in it. Talk with your clinician before starting new exercise or supplements, particularly if you take blood thinners or have osteoporosis, and respect the pressure ceilings above, which differ for manual tools and powered devices.